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Anatomical insights into the superior longitudinal system from integrative in-vivo and ex-vivo mapping



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Charting the organization of white matter (WM) pathways is essential for understanding the functioning of the human brain. This study provides a comprehensive, anatomically enhanced characterization of the superior longitudinal system (SLS) by integrating in-vivo tractography with ex-vivo dissection within the same radiological space. Using a data-driven cortex-to-cortex pairing approach leveraging gyral-sulcal macroanatomical landmarks, we reconstructed the dorsal associative connectivity of the frontal cortex in 39 healthy participants. We identified 45 SLS components, of which (i) 22 were validated and refined through ex-vivo dissection, (ii) 17 were deemed anatomically plausible despite lacking ex-vivo confirmation, and (iii) 6 were classified as anatomically implausible. The anatomical description of plausible sub-SLS templates revealed fundamental organizational principles of the system: (i) a medio-lateral and dorso-ventral hierarchy, where dorsal regions connect dorsally and ventral regions ventrally, and (ii) a depth-dependent organization, with shorter, superficial fibers linking proximal areas and longer, deeper fibers connecting distal regions. Pearson's correlation confirmed a significant positive relationship between streamline length and distance from the cortex ($r = 0.689$, $p < 0.001$). This study emphasizes the need for a distributed and integrated understanding of brain connectivity beyond classical bundle definition, and provides a robust anatomical foundation for future population-based WM atlases using bundle-specific tractography.

The Superior Longitudinal System (SLS) encompasses white matter (WM) fibers connecting the frontal lobe to the other cortices of the same hemisphere by coursing above the insula¹. It is traditionally described through the characterization of two main representative WM bundles: the arcuate (AF) and superior longitudinal (SLF) fasciculi², with fronto-temporal fibers classified as part of the AF and fronto-parietal fibers belonging to the SLF³. Despite being extensively studied⁴, there remains no consensus about the anatomical extent, course, and terminations of the AF and the SLF, and the literature surrounding these WM pathways is notably complex^{2,5}.

Historically, the AF and the SLF - labels originally used interchangeably to refer to the same WM pathway - were described as a single group of fibers originating in the frontal cortex and fanning into the parietal, temporal, and occipital cortices^{6,7}. As Joseph Klingler drastically advanced the technique of ex-vivo dissection, granting a better isolation of WM fibers⁸, and diffusion-weighted imaging (DWI)-based tractography ushered in their in-vivo study⁹, the characterization of the AF/SLF complex soon became particularly detailed. This led to a finer delineation of the termination territories of these connections and to the identification of sub-components based on

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topological features (e.g.,^{10–13}). The rapid expansion of the literature on the AF/SLF in the first two decades of the 21st Century has resulted in different concurrent interpretations of the original nomenclature, presenting two complex scenarios. First, identical anatomical connections defined by their cortical terminations have been labeled differently across studies. For instance, the connection between the precentral gyrus (PrCG) and the middle temporal gyrus (MTG) can be variably found in the literature as “AF”^{14–16}, “SLF temporofrontal segment”¹⁷, “long segment/direct component of the AF”¹⁰, “MTG pathway”¹¹, or “outer/dorsal pathway”¹⁸. Conversely, the same label has been used to denote distinct wiring patterns. For example, “AF” has been used to describe a fiber bundle connecting the middle frontal gyrus (MFG) to the superior temporal gyrus (STG)¹⁹, as well as a fiber bundle connecting the PrCG and inferior frontal gyrus (IFG) to the MTG and STG²⁰, and another fiber bundle connecting the PrCG, IFG, and MFG to the STG, MTG and inferior temporal gyrus (ITG)¹⁴. This variability makes comparisons between studies virtually impossible, undermining a unified understanding of the anatomical definition of these WM pathways, and hindering the interpretation of findings in both basic and clinical neuroscience.

Even when discrepancies related to nomenclature are resolved, the structural wiring of the AF/SLF complex is not univocally agreed upon, and anatomical descriptions vary significantly among studies². This variability mainly derives from the fact that descriptions of the cortical terminations of a bundle are bound to the specific aim of each study and rely on the assumption of prior anatomical definitions for bundle segmentation^{19,21–23}. Moreover, using different tractography models and algorithms adds further variability to the reconstruction of WM pathways²⁴. Collectively, these factors contribute to incomplete or inconsistent representations of the overall structural wiring of WM systems. To date, no study has assessed the complete extension of the SLS.

The present study aims to provide a comprehensive and anatomically evaluated tractography reconstruction of the SLS, representing the entire dorsal longitudinal connectivity of the frontal lobe with the other ipsilateral cortices. By considering WM fiber pathways as systems rather than discrete bundles¹, we refrain from assuming prior anatomical definitions and account for the full extent of the dorsal association fibers of the frontal cortex through a data-driven approach. In this framework, the architecture of the SLS is defined by patterns of connectivity among established macro-anatomical landmarks, namely sulco-gyral features of the brain. We contrast each connectivity pattern - referred to as sub-SLS (i.e., SLS subcomponent)—with ex-vivo Klingler microdissection data⁸. This multimodal in-vivo/ex-vivo comparative approach has previously been applied to the study of the AF and SLF (e.g.,^{13,17}), and is regarded as the gold standard for reconstructing and validating the WM connections of the human brain. Herein, we further enhance the contribution of anatomical data from ex-vivo dissection by employing the BraDiPho framework (<https://bradipho.eu>)²⁵, merging in-vivo tractography with 3D photogrammetry reconstructions of ex-vivo dissections in the same radiological space, thus providing a one-to-one, precise, and rigorous evaluation of tractography data. The template of each sub-SLS will become a source of anatomically relevant priors to be injected into the Bundle-Specific Tractography (BST) approach²⁶ for an unprecedented atlasing of the wiring of the SLS.

Results

We performed constrained spherical deconvolution (CSD) and particle-filtering tractography (PFT) with anatomical priors^{27,28} on DWI data of 39 healthy participants of the BIL&GIN database (Fig. 1A)²⁹. All tractograms were registered to MNI space³⁰, and the SLS was extracted for each individual leveraging the regions of interest (ROIs) of the JHU cortical atlas (Fig. 1B)³¹. Individual SLS were then concatenated. The N39 SLS was then parceled into sub-SLS using a cortex-to-cortex pairing approach, in which each frontal gyrus was systematically paired with every other gyrus of the same hemisphere as defined by the JHU template (Fig. 1C). sub-SLS from the right hemisphere were flipped and concatenated with their left hemisphere homologous (Fig. 1D). Out of the 119 possible ROI combinations,

the cortex-to-cortex pairing approach led to the extraction of 96 non-empty sub-SLS tractograms. Of these, 45 contained more than 200 streamlines and were subsequently processed with QuickBundlesX clustering for the initial exclusion of implausible streamlines based on their geometrical features (Fig. 1D)³². These sub-SLS revealed a defined wiring of connections linking the 4 frontal gyri of the lateral convexity (i.e., IFG, MFG, superior frontal gyrus (SFG), and PrCG) to several regions covering the lateral surface of the parietal (i.e., angular gyrus (AG), postcentral gyrus (PoCG), superior parietal gyrus (SPG) and supramarginal gyrus (SMG)), temporal (i.e., ITG, MTG, STG and temporal pole (T-pole)) and occipital cortices (i.e., inferior (IOG) and middle occipital gyri (MOG)), and a restricted section of the basal temporal cortex (i.e., fusiform gyrus (FuG)) (Fig. 2). Connections involving the FuG, IOG, and T-pole were null when paired with the SFG. On the medial surface, connectivity was confined to the precuneus (PrCu). Conversely, 74 ROI combinations, consisting of 33 empty tractograms (Fig. 2, black circles) and 41 with less than 200 streamlines in both hemispheres (Fig. 2, gray circles), were deemed non-viable. They included the entire connectivity of the fronto-orbital region (lateral fronto-orbital gyrus (LFOG), middle fronto-orbital gyrus (MFOG), and rectus gyrus (RG)) as well as connections to the cuneus (Cu), lingual gyrus (LG), parahippocampal gyrus (PHG), entorhinal cortex (ENT), and superior occipital gyrus (SOG). Additionally, as noted earlier, connections between the SFG and the IOG, FuG, and T-pole were null.

Sub-SLS tractography templates description

This section is dedicated to the anatomical description of the 45 sub-SLS tractography templates obtained with the cortex-to-cortex pairing approach and after the removal of streamlines classified as implausible based on their geometrical features, as represented in Fig. 2. For this description, templates are grouped based on the frontal gyrus from which they originate.

Inferior frontal gyrus. Connections between the IFG and the inferior parietal lobule (IPL), namely AG and SMG, primarily originate from *pars opercularis* (pOp) and *pars triangularis* (pTri), with only a few streamlines reaching the dorsal and posterior *pars orbitalis* (pOrb). These connections course longitudinally within the deep WM of the central region of the hemisphere, running parallel the Sylvian fissure before resurfacing laterally to reach their parietal target. Connections to the AG mostly terminate in its anterior part, while those to the SMG are more evenly distributed. The IFG-PoCG template follows a similar but shorter course, paralleling the Sylvian fissure. For this connection, terminations in the IFG are more densely distributed in pOp and become sparser in pTri, with no streamline reaching pOrb. Posterior terminations cover only the most ventral part of the PoCG. The two templates representing connections to the dorso-medial parietal cortex, namely IFG-SPG and IFG-PrCu, follow a similar anatomical course to the connections described above up to the level of the PoCG, where they shift dorsally to reach their respective targets. These connections arise primarily from pOp, with only a few streamlines reaching pTri, particularly for the IFG-PrCu template, and reach anterior SPG and anterior PrCu. Connections between the IFG and the temporal lobe follow the same longitudinal course as the IFG-parietal connections up to the deep WM at the level of the AG. At this point, streamlines arch around the posterior ramus of the Sylvian fissure, then proceed ventrally with a transverse orientation. After entering the deep WM of the temporal lobe, they proceed towards their target temporal gyrus and start fanning to resurface laterally. While their fanning enlarges both anteriorly and posteriorly for connections to the ITG, MTG, STG, and FuG, connections to the T-pole follow a postero-anterior direction as they traverse the deep WM of the dorsal temporal lobe, paralleling the Sylvian fissure ventrally. Connections reaching the STG and the MTG arise uniformly from both pOp and pTri, with fewer streamlines originating from pOrb, while connections with the FuG, ITG, and T-pole almost exclusively originate from pOp. Posteriorly, connections to the FuG cover only the mid-segment of the ROI. While connections to the ITG, MTG, and STG distribute equally along the posterior ⅓ of the whole gyri, connections to the T-pole

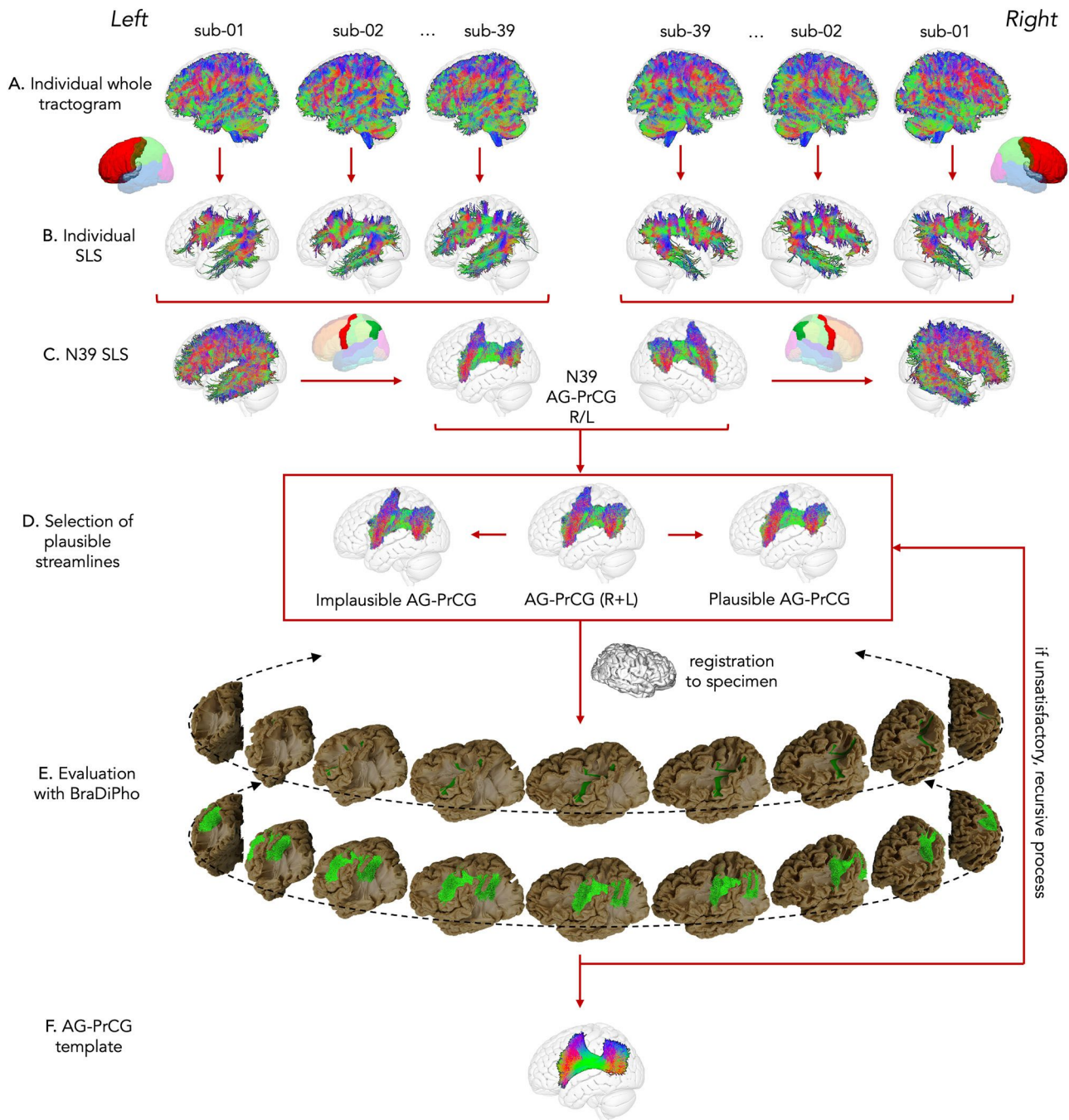


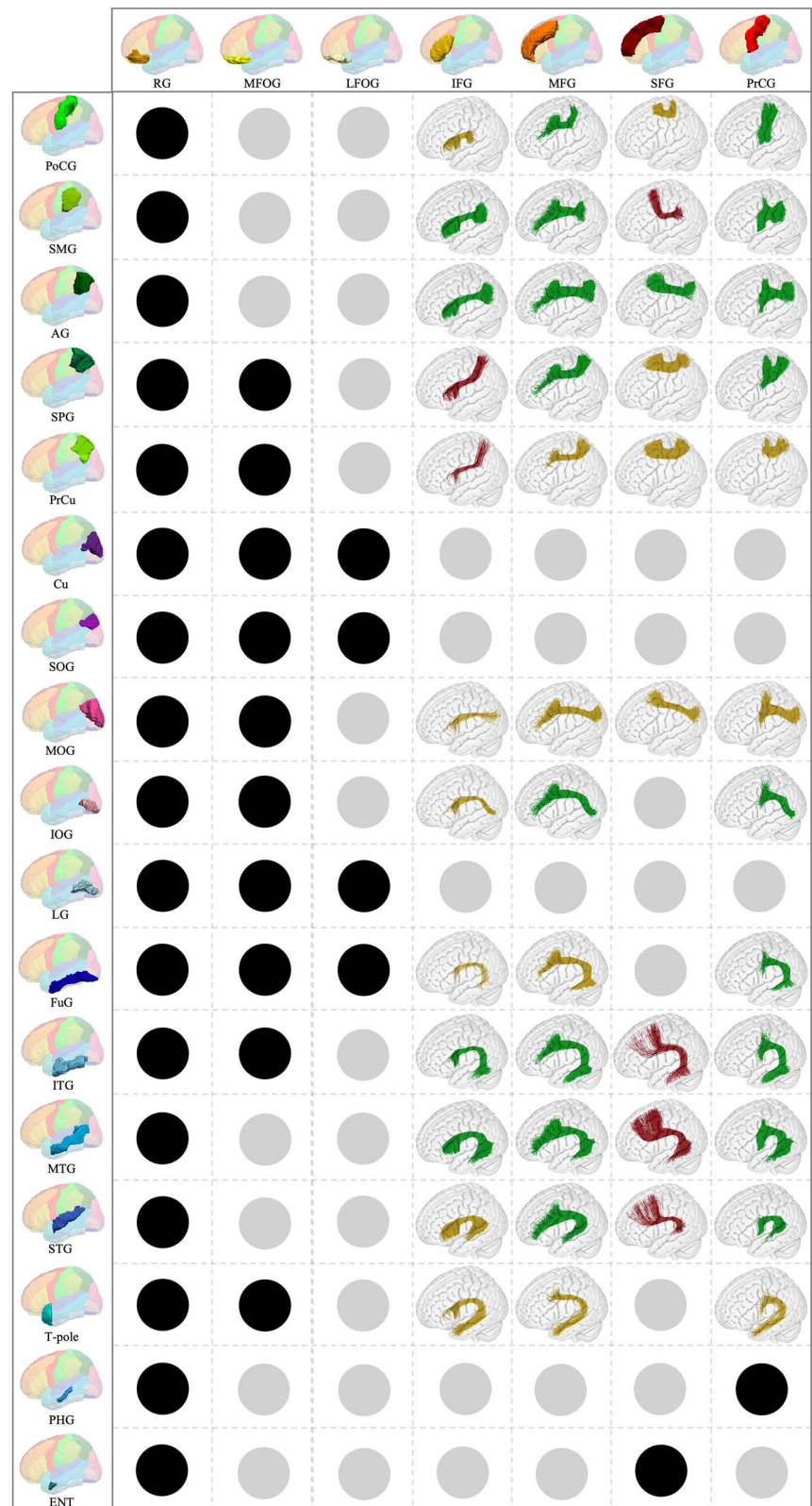
Fig. 1 | Schematic representation of the pipeline for creating sub-SLS templates. Starting from the whole-brain tractograms of 39 individuals in MNI space (A), individual SLS tractograms are extracted (B). Individual SLS tractograms are concatenated, and the cortex-to-cortex pairing approach is applied to parcellate the system into sub-SLS (C). sub-SLS from the right hemisphere are flipped to the left hemisphere and concatenated to their contralateral homologs (C). Broad clustering is

performed, and streamlines of each sub-SLS are classified as plausible or implausible based on geometrical features (D). Filtered sub-SLS undergo anatomical evaluation using the BraDiPho framework (E), followed by smoothing (F). If anatomical evaluation identifies gaps in spatial coverage, the process from D to F is repeated, with finer clustering and refinement based on the information from ex-vivo dissection. AG angular gyrus, PrCG precentral gyrus.

correspond to the anterior continuation of these connections. For what concerns connections with the occipital cortex, the IFG-MOG template follows a course similar to IFG-parietal connections, while the IFG-IOG template is more akin to IFG-temporal connections. Indeed, the IFG-MOG template has an overall longitudinal orientation, whereas IFG-IOG fibers arch around the Sylvian fissure, spanning the boundary between the temporal and occipital cortices before fanning out posteriorly to reach the anterior portion of IOG.

Middle frontal gyrus. Connections between the MFG and the IPL originate from the posterior $\frac{2}{3}$ of the MFG ROI. After leaving the gyrus, streamlines travel posteriorly within the deep WM of the central region, maintaining a longitudinal orientation along the z-axis (streamlines course at the same height from the posterior end of the MFG to the AG and SMG). Streamlines resurface laterally at the level of the gyrus they terminate into. Posterior terminations are homogeneously distributed across both the AG and SMG ROIs. Templates of connections from the MFG to the PoCG, SPG, and PrCu exhibit a more compact course

Fig. 2 | Overview of the sub-SLS obtained through the process outlined in Fig. 1. It shows the 45 viable gyral combinations identified with the cortex-to-cortex pairing approach, along with combinations that resulted in empty tractograms (black circles) or tractograms containing less than 200 streamlines each (gray circles). Viable sub-SLS were classified as *anatomically plausible and validated* (green), *anatomically plausible but not validated* (yellow), and *anatomically implausible* (red) based on their correspondence with ex-vivo dissection data and adherence to anatomical principles of fibers distribution. AG angular gyrus, Cu cuneus, ENT entorhinal cortex, FuG fusiform gyrus, IFG inferior frontal gyrus, IOG inferior occipital gyrus, ITG inferior temporal gyrus, LFOG lateral fronto-orbital gyrus, LG lingual gyrus, MFG middle frontal gyrus, MFOG medial fronto-orbital gyrus, MOG middle occipital gyrus, MTG middle temporal gyrus, PHG parahippocampal gyrus, PoCG postcentral gyrus, PrCG precentral gyrus, PrCu precuneus, SFG superior frontal gyrus, SMG supramarginal gyrus, SOG superior occipital gyrus, SPG superior parietal gyrus, STG superior temporal gyrus, RG rectus gyrus, T-pole temporal pole.



compared to connections to the IPL, and course slightly dorsal to them. Streamlines connecting to the SPG display more anterior terminations in MFG, covering the posterior $\frac{2}{3}$ of the ROI similarly to the IPL connections. In contrast, streamlines to the PoCG and PrCu tend to terminate more posteriorly in the MFG, within the posterior third of the region. Streamlines of the MFG-PoCG template resurface laterally after passing

the level of the central sulcus, with posterior terminations covering only the middle third and the inferior part of the dorsal third of the PoCG ROI. Connections from the MFG to the SPG and PrCu shift slightly vertically after passing the level of the postcentral sulcus, reaching the anterior part of the SPG and the entire PrCu ROI. Connections between the MFG and the ITG, MTG, and STG are more widespread, covering the posterior $\frac{2}{3}$

of the MFG ROI. In contrast, connections to the FuG and the T-pole tend to arise more posteriorly in MFG, with the MFG-T-pole template covering only the very posterior aspect of the frontal ROI. After coursing longitudinally in the deep WM of the frontal and parietal lobes, MFG-temporal streamlines arch around the Sylvian fissure at the level of the IPL and enter the temporal cortex along a transverse course. Connections to the FuG terminate in the mid-segment of the FuG ROI, while connections to the ITG, MTG, and STG span the posterior $\frac{2}{3}$ of the lateral temporal gyri. Connections to the T-pole traverse the deep WM of the superior temporal lobe and extend anteriorly. For what concerns connections with the occipital cortex, the MFG-MOG template resembles in its course MFG-parietal connections, while MFG-IOG conforms with MFG-temporal connections, arching around the Sylvian fissure and coursing ventrally with a transverse orientation before fanning posteriorly to reach the anterior end of the IOG ROI. The MFG-MOG template terminates in the anterior MOG.

Superior frontal gyrus. Connections between the SFG and the parietal lobe originate solely from the posterior third of the SFG. Templates reaching the PoCG, SPG, PrCu, and AG follow a longitudinal course through the deep WM of the dorsal central region. Contrarily, connections reaching the SMG leave the SFG ventrally, follow a main transverse orientation up to the level of the deep WM below the ventral PrCG and abruptly turn longitudinally before traveling posteriorly to the anterior SMG. Terminations in the PoCG cover only the most dorsal portion of the ROI, while connections to the SPG and PrCu cover the entire gyri. The AG is mainly covered in its antero-dorsal part. Connectivity between the SFG and the occipital cortex is represented by the sole SFG-MOG template. It resembles the SFG-AG template in its course, but extends further posteriorly to the anterior MOG. SFG-temporal connections involve only the posterior halves of the ITG, MTG, and STG. These streamlines leave the posterior half of the SFG ventrally and traverse the frontal lobe with a transverse orientation, up to the level of the dorsal IFG. Here, they abruptly switch course to a longitudinal orientation, reaching the deep WM of the IPL before arching into the temporal lobe. Streamlines fan laterally, both posteriorly and anteriorly, at the level of their respective temporal gyrus.

Precentral gyrus. The PrCG-PoCG template represents the only pattern of connectivity between adjacent gyri of the SLS. It consists of U-shaped fibers that span the entire extent of both gyri. Connections between the PrCG and the IPL arise from the ventral half of the PrCG ROI. After coursing briefly through the deep WM of the central region, streamlines resurface laterally, terminating homogeneously along the entire cortical surface of both the AG and SMG. The fanning of these streamlines is larger anteriorly to accommodate for the involvement of the PrCG ROI, which develops primarily along the z-axis. Connections between the PrCG and the SPG or the PrCu originate more dorsally in the PrCG than those to the IPL. Streamlines connecting to the SPG originate from the dorsal half of the PrCG, while those reaching the PrCu originate from the most dorsal third of this region. Streamlines of the PrCG-SPG template group in the deep WM below the PoCG and then turn slightly vertically to reach the anterior half of the SPG, while PrCG-PrCu connections remain dorsal and cover the anterior $\frac{2}{3}$ of PrCu. Connections between the PrCG and the temporal lobe arise from the ventral half of the PrCG. Streamlines connecting to the ITG and the MTG are uniformly distributed along this portion of the PrCG ROI, those to the T-pole and the FuG arise more dorsally compared to the others, and those to STG seed only in the most ventral end of the region. Streamlines of the PrCG-temporal templates travel longitudinally up to the deep WM below the IPL, arch around the Sylvian fissure, and follow a transverse course until they reach the deep WM of their target gyrus. Connections to the FuG terminate in the mid-segment of the ROI, while those to the ITG, MTG, and STG cover the posterior $\frac{2}{3}$ of the respective gyri. Connections to the T-pole extend more anteriorly, paralleling the Sylvian fissure ventrally, and cover the anterior

third of the temporal lobe. As previously described for IFG and MFG, the PrCG-MOG template follows a similar course to PrCG-parietal connections, while streamlines of the PrCG-IOG template arch into the temporal lobe, crossing the boundary between temporal and occipital cortices, and fan posteriorly.

Sub-SLS ex-vivo dissection

Layer-by-layer Klingler microdissection of 6 left and 4 right hemispheres resulted in the identification of 24 dissection layers (henceforth referred to as epochs - abbreviated as “epo” when referring to a specific dissection layer) representing connections of the SLS. In some specimens, (abbreviated as “spc” when referring to a specific specimen) both fronto-parietal and fronto-temporal connectivity were exposed while proceeding with a latero-medial dissection approach (e.g., Fig. 3, spc-01, spc-06, spc-07, spc-10, spc-12, spc-15, and spc-19). Occasionally, fronto-parietal connectivity could be exposed on two subsequent epochs of the same specimen: a lateral one revealing more superficial and shorter fibers, and a medial one revealing deeper and longer ones (e.g., Fig. 3, spc-02 and spc-07). 3D digital models of each epoch were reconstructed with photogrammetry and aligned to the radiological space of each specimen^{25,33}. A detailed anatomical analysis of the trajectory of the dorsal fibers leaving the frontal cortex and connecting to any other cortex of the same hemisphere was carried out on these models. Each sub-SLS connection visible on the texture of the models was manually annotated with the CloudCompare “Segment Tool” following a cortex-to-cortex pairing approach, compatible with the one adopted for the parcellation of tractography data. The identified sub-SLS annotated on the different epochs are listed and represented in Fig. 3.

Cortex-to-cortex annotation of the SLS connections on the specimens resulted in the identification of 22 sub-SLS, all of which were also previously reconstructed with tractography and represented by sub-SLS templates. These annotations depict connections between the IFG with the IPL (i.e., AG and SMG) as well as with the ITG and MTG. Connections linking the MFG and the PrCG to the lateral aspect of the parietal (i.e., AG, SMG, PoCG, and SPG) and temporal cortices (i.e., STG, MTG, and ITG) were also traced, alongside those to the IOG. The SFG-AG connection was the only sub-SLS identified from the SFG. Fibers reaching the FuG were visible just in one specimen, where they connected to the PrCG.

While the manual annotation of sub-SLS connections on the dissection models confirmed the existence of these connectivity patterns, the concurrent analysis of all the sub-SLS annotations across epochs revealed several organizational principles of the WM fibers of the SLS. Shorter fibers were found to course more superficially, while longer fibers ran more deeply. Specifically, fronto-parietal connections, being shorter by nature, were found on more superficial epochs compared to fronto-temporal fibers, which required the removal of overlying tissue (e.g., spc-01 and spc-02 epo-06 vs. epo-07 in Fig. 3). While this observation aligns with the classical description of the SLF/AF complex³⁴, we also noted that this organizational principle holds within individual connections defined by specific termination territories. For instance, MFG-AG connection in spc-02 was identified on two consecutive epochs (see Fig. 3, spc-02, epo-05 and epo-06 in orange). On epo-05, MFG-AG appears as a short connection involving only posterior MFG, while on epo-06, deeper fibers extend to reach anterior MFG. This latero-medial organizational pattern contributes to a layered architecture of WM fibers of the SLS, which also respects a ventro-dorsal organization. Ventral regions connect to other ventral regions through fibers coursing ventrally, while dorsal regions are connected to other dorsal regions through fibers coursing dorsally. This is particularly evident in fronto-parietal epochs. For instance, in epo-06 of spc-01 connections between the IFG and the AG course more ventrally compared to those between the MFG and the AG. Conversely, connections between the PrCG and the AG remain dorsal if originating from dorsal PrCG and ventral if arising from ventral PrCG. As seen in epo-07 of spc-02, this organizational layout also applies to fronto-temporal connections. In the case of PrCG-ITG fibers, those arising from the dorsal part of the PrCG after the splitting of the gyrus remain more external as they arch into the temporal lobe (i.e., farther from the posterior



Fig. 3 | Layer-by-layer Klingler microdissection of 10 hemispheres, with the identification of sub-SLS on photogrammetric models. The annotation of the sub-SLS across 24 epochs according to the cortex-to-cortex pairing approach revealed the anatomical features of 22 sub-SLS previously identified with tractography. SPC specimen, EPO epoch. AG angular gyrus, FuG fusiform gyrus, IFG inferior frontal

gyrus, IOG inferior occipital gyrus, ITG inferior temporal gyrus, MFG middle frontal gyrus, MTG middle temporal gyrus, PoCG postcentral gyrus, PrCG pre-central gyrus, SFG superior frontal gyrus, SMG supramarginal gyrus, SPG superior parietal gyrus, STG superior temporal gyrus.

end of the Sylvian fissure) compared to fibers originating from the ventral part of the PrCG, which run closer to the Sylvian fissure. Thus, the ventro-dorsal organization in fronto-temporal connections translates into a mirrored pattern relative to the Sylvian fissure. Specifically, connections more ventral in the frontal cortex tend to connect to more dorsal regions in the temporal cortex, following the shape of the Sylvian fissure closely. Interestingly, as fronto-temporal fibers of the SLS arise more dorsally in the frontal cortex and thus course farther from the Sylvian fissure, the shift from a longitudinal course in the frontal and parietal cortices to a transverse course in the temporal cortex becomes less steep. Eventually, this results in an inversion in the fanning of temporal fibers - from more anteriorly directed for internal connections to more posteriorly directed for external connections. This trend culminates in fibers bending posteriorly to reach the occipital cortex, as shown in epo-05 of spc-16.

Sub-SLS templates anatomical evaluation with BraDiPho

All the 45 sub-SLS tractography templates were non-linearly registered to the radiological space of the 10 3D models of ex-vivo dissection for

anatomical evaluation (Fig. 1E). Of these, 22 sub-SLS templates obtained with tractography were matched with their corresponding anatomical annotations from ex-vivo dissection, as described in the previous section. Quantitative overlap scores between tractography reconstructions and the corresponding manual annotations on the photogrammetric models of ex-vivo dissection are reported in Supplementary Data 1 and visualized in Supplementary Fig. 1. Across all matched sub-SLS templates, the average overlap between modalities was $88.15\% \pm 11.60\%$ (mean $\pm$ SD; min 67.8%; max 100%). It is important to note that a degree of spatial mismatch may be expected, as tractography is registered to the T1 of the specimens before starting the dissection and minor shifts in tissue conformation occur during the dissection process, particularly as tissues relax following removal of overlying structures.

Based on the representation of the sub-SLS in the ex-vivo dissection models, the tractography templates were further refined. Initially, the selection of plausible streamlines was made without reference to the anatomical features of the sub-SLS, relying instead on plausibility judgment based on the geometrical characteristics of the streamlines. Moreover, this

process involved broad clusters encompassing a wide array of streamlines, sometimes intermingling both plausible and implausible ones, possibly leading to the rejection of some plausible streamlines along with implausible ones. While rejecting some plausible streamlines would typically not be problematic in most cases due to the redundancy of tractography, with the correction for density bias in creation of the templates it could result in the loss of relevant reconstructions. Therefore, once anatomical information was obtained through the one-to-one comparison of in-vivo tractography and ex-vivo dissection data, the selection of plausible streamlines was re-performed in Tractome. This allowed for a more selective, connection-specific filtering of implausible streamlines^{35,36}. The re-evaluation of the sub-SLS templates resulted in a more comprehensive and anatomically informed representation of WM connections, with improved coverage of their anatomical extent as suggested by the dissection data. These *anatomically plausible and validated* sub-SLS templates are shown in green in Fig. 1. A graphical summary of the anatomical evaluation process with BraDiPho is provided for each template in Supplementary Fig. 1.

The remaining 23 sub-SLS tractography templates were not found in our dissection data. These templates were classified as *anatomically plausible but not validated* or *anatomically implausible* based on the information gathered through dissection. The *anatomically plausible but not validated* templates (Fig. 2, yellow templates) represent connections that were reconstructed with tractography but could not be confirmed in our dissections, although their anatomical course remains consistent with *anatomically plausible and validated* connections. For instance, the existence of very medial connections, such as those from the MFG/PrCG/SFG to the PrCu, between SFG and dorsal PoCG, and between SFG and SPG, could not be proven, but they could not be excluded either. Indeed, these connections could not be exposed by our latero-medial dissection approach, which was not optimal for exposing such WM pathways. Similarly, connections to the MOG were not found in our dissections, as the dissection of purely longitudinal fibers stopped at the level of the AG. Dissection of the temporal cortex also did not extend to the T-pole, and rarely revealed connections to the basal surface of the hemisphere. The only specimen showing connections to the FuG (i.e., PrCG-FuG, see Fig. 3) was spc-06. Indeed, its unique anatomical configuration showing a “bulge” of the FuG interposing between the ITG and the IOG on the ventral lateral aspect of the convexity allowed the exposure of fibers connecting to this region through a latero-medial dissection approach. Additionally, connections with the IFG were only observed in 3 specimens, leaving anatomically plausible connections such as IFG-PoCG, IFG-STG, and IFG-IOG unproven.

On the other hand, we classified as *anatomically implausible* those sub-SLS templates showing abrupt changes of direction (Fig. 2, red templates). For instance, sub-SLS templates between the SFG and the temporal cortex (ITG, MTG, and STG) leave the posterior half of the SFG ventrally and traverse the posterior frontal cortex transversally up to the IFG before joining the other fronto-temporal connections of the SLS. Similarly, connections of the SFG-SMG template leave the SFG ventrally, travel vertically up to the level of the IFG, and then abruptly bend longitudinally toward the SMG. A similar but opposed shift in direction is seen in the IFG-SPG and IFG-PrCu templates, which travel longitudinally from the IFG up to the postcentral sulcus, before shifting vertically to reach the dorsal and medial parietal cortices. No such abrupt changes in direction were observed in the ex-vivo dissection of the SLS, therefore, these templates were classified as anatomically implausible. These directional shifts are likely caused by the intersection of longitudinal fibers with the anterior transverse system in the frontal lobe or with projection fibers of the pyramidal tract in the parietal lobe, leading the tractography algorithm to follow undesired directions.

sub-SLS templates replication study

The replication analysis of the sub-SLS tractography templates carried out on the three HCP test-retest tractography datasets distributed on brainlife.io³⁷ with the Fast Streamline Search approach³⁸ revealed a good agreement across tractography reconstructions derived from two different DWI acquisition protocols (i.e., BIL&GIN vs. HCP). Overall, we achieved

0.77 ± 0.27 average wDSC (i.e., dice coefficient on voxels weighted for streamline density) for *test* and *test_run_2* and 0.78 ± 0.26 for *retest*. When considering only *anatomically plausible and validated templates*, mean wDSC spiked to 0.92 ± 0.08 for all HCP datasets, while it attested to 0.72 ± 0.29 , 0.71 ± 0.3 , and 0.71 ± 0.31 for *test*, *test_run_2*, and *retest* respectively, when considering *anatomically plausible but not validated* templates. On the other hand, wDSC dropped to 0.39 ± 0.22 , 0.37 ± 0.19 , and 0.43 ± 0.18 when considering sub-SLS templates deemed *anatomically implausible*. All measurements, including further streamline- and voxel-based similarity measurements on top of wDSC (i.e., bundle adjacency, dice coefficient, density correlation, overlap and overreach) are available in Supplementary Data 2, also reporting mean and standard deviations calculated considering all the templates together as well as filtered according to the classification of the templates (*anatomically plausible and validated* in green, *anatomically plausible but not validated* in yellow, and *anatomically implausible* in red). Single tractography reconstructions and voxel masks showing the overlap between the sub-SLS provided in the present work and sub-SLS extracted from each HCP dataset are displayed in Supplementary Fig. 2 with the respective wDSC. Of note, while on average wDSC dropped for *anatomically implausible* templates, the visualization provided in Supplementary Fig. 2 shows that implausible streamlines following the exact same course of the ones extracted on the N39 BIL&GIN dataset could also be found in the three HCP datasets. This reinforces the importance of relying on direct anatomical explorations to make tractography-based inferences on the architecture of WM pathways, independently from the characteristic of DWI and tractography.

sub-SLS templates characterization

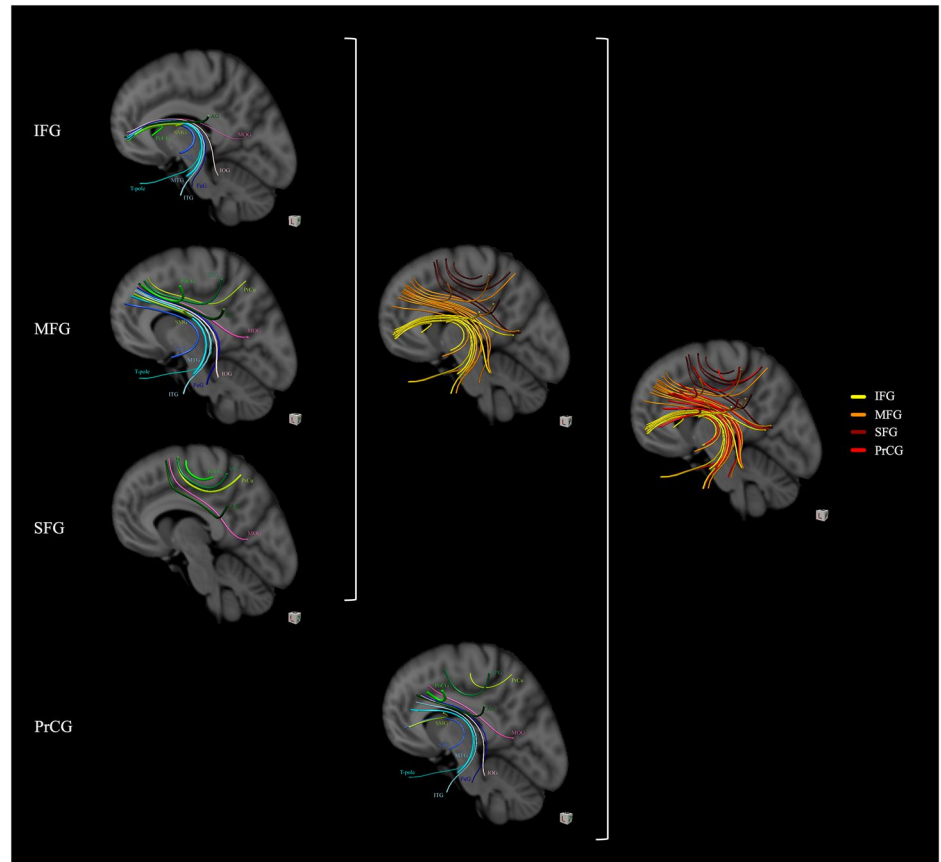
A numerical description of each *anatomically plausible* sub-SLS tractography template—both *validated* and *not validated* (Fig. 2, green and yellow templates, respectively)—is provided through shape analysis (Supplementary Data 3)³⁹. Additionally, we provide the mean coordinates of both anterior and posterior terminations for each of these templates in MNI space, with “anterior” indicating frontal terminations (Supplementary Data 4).

sub-SLS templates topological organization

The study of the SLS wiring through ex-vivo dissection and the annotation of different sub-SLS on photogrammetric models revealed key organizational principles of the WM fibers of this system. We analyzed the sub-SLS tractography templates to test for the occurrence of the same medio-lateral and dorso-ventral patterns of fiber distribution identified in the dissection study. The combined visualization of centroids of the sub-SLS templates originating in the IFG, MFG, and SFG shows that IFG connections start, course, and terminate more ventrally compared to MFG connections, which in turn are more ventral compared to SFG connections (Fig. 4, middle column, top row, in yellow, orange, and bordeaux, respectively). As previously described in the dissection data, this ordered layout is respected when streamlines bend into the temporal cortex, and defines their distance from the Sylvian fissure when arching around it (i.e., the more dorsal in the frontal cortex, the farther they course from the Sylvian fissure). Connections originating in the PrCG, which parallels the central sulcus and spans the most posterior part of the frontal lobe with a dorso-ventral configuration, overlay to those arising from the other frontal gyri (Fig. 4, third column).

The dorso-ventral organization of fibers observed by considering all the frontal gyri together is respected also at the level of the single gyrus (Fig. 4, icons dedicated to IFG, MFG, SFG, and PrCG in the first and second columns). Connections arising in the PrCG provide an illustrative example. As it can be observed in Fig. 4, (second column, PrCG icon) fronto-parietal connections, depicted in green, are organized according to a ventro-dorsal layout, with a sequential organization orderly displaying, from ventral to dorsal, PrCG-SMG, PrCG-AG, PrCG-PoCG (whose centroid lies in the middle of the gyrus, as the connection spans the whole extent of both PrCG and PoCG), PoCG-SPG, and PrCG-PrCu. This organization respects the arrangement of cortical gyri in the parietal cortex, and drives the distribution

Fig. 4 | Centroids reconstruction of the anatomically plausible sub-SLS templates. Centroids representing the SLS connectivity for each frontal gyrus are shown in the first (IFG, MFG, SFG) and second columns (PrCG, bottom row), with parietal, temporal, and occipital connections in green, blue, and pink, respectively. A concatenation of the centroids representing sub-SLS templates originating from the IFG (yellow), MFG (orange), and SFG (bordeaux) is provided in the second column, top row. The third column icon displays all anatomically plausible sub-SLS centroids together with IFG connections in yellow, MFG connections in orange, SFG connections in bordeaux and PrCG connections in red). AG angular gyrus, FuG fusiform gyrus, IFG inferior frontal gyrus, IOG inferior occipital gyrus, ITG inferior temporal gyrus, MFG middle frontal gyrus, MOG middle occipital gyrus, MTG middle temporal gyrus, PoCG postcentral gyrus, PrCG precentral gyrus, PrCu precuneus, SFG superior frontal gyrus, SMG supramarginal gyrus, SOG superior occipital gyrus, SPG superior parietal gyrus, STG superior temporal gyrus, T-pole temporal pole.



of the fibers. The same principle applies to connections between the PrCG and both the lateral aspect of the temporal cortex and the occipital lobe. Connections from the PrCG to the STG terminate more ventrally in the PrCG compared to those reaching the MTG, which, in turn, are more ventral than those reaching the ITG. Accordingly, as we proceed from ventral to dorsal in the PrCG - and inversely from dorsal to ventral in the temporal cortex -, streamlines shift from coursing close to the Sylvian fissure to following a more distant trajectory. Similarly, connections to the IOG arise more ventrally in the PrCG compared to those reaching the MOG, and this dorso-ventral organization persists along their entire course. Connections to the T-pole and to the FuG demonstrate a similar organization, forming an additional deeper layer. Indeed, as seen in all the icons of Fig. 4 dedicated to each single gyrus, fronto-parietal connections, depicted in green, are the most superficial, followed by connections to the lateral temporal cortex and to the occipital lobe, which are themselves more superficial compared to connections reaching the anterior temporal lobe (*i.e.*, T-pole) and the basal temporal cortex (*i.e.*, FuG).

This medio-lateral layering reflects the length of streamlines, with shorter connections coursing more superficially and longer connections traveling deeper. This principle is shown in Fig. 5A, B, where streamlines are color-coded based on their distance from the cortex. More superficial fibers (yellow) cover shorter distances, while longer fibers (dark red), travel more deeply. Supplementary Fig. 3 provides the same color-coded representation for each sub-SLS template, further demonstrating that this organizational principle applies broadly across the SLS, even when considering only part of the system. Statistical analysis revealed a significant positive correlation between the length of each streamline and its distance from the cortex (Pearson's $r = 0.689$, $p < 0.001$) (Fig. 5C).

Discussion

In the era of connectomics, having a precise, validated, and broadly shareable knowledge of the WM anatomy is crucial for interpreting findings in

both basic and clinical neuroscience. Traditional frameworks for describing WM anatomy focus on discrete bundles, such as the SLF and the AF, as well as the inferior fronto-occipital fasciculus (IFOF), the uncinate fasciculus (UF), the frontal aslant tract (FAT), as representatives of the whole associative connectivity of the brain⁴. While these bundles represent key patterns of connectivity between ipsilateral cortices by summarizing the general topology of WM fibers¹, their historical characterization has shaped ex-vivo dissection protocols and tractography segmentation approaches, often limiting our understanding of WM connectivity^{36,40–44}. This reductionist view does not capture the true complexity of the structural architecture of the brain, which requires more distributed and integrated systems of connections to support the emergence of cognitive functions⁴⁵.

The present work provides a comprehensive anatomically enhanced characterization of the SLS. We reconstructed the wiring diagram of the whole dorsal associative connectivity of the frontal cortex without imposing predefined anatomical boundaries to the extension of this pathway but the sulci that delimit the frontal convolutions. We selected fibers of the SLS from whole-brain tractograms according to their gross anatomical classification¹ and we parceled the system into sub-components through a cortex-to-cortex pairing approach. Specifically, streamlines with one ending in the frontal cortex and connecting to any other cortex of the ipsilateral hemisphere by coursing above the insula were retained as part of the SLS. Macroanatomical sulco-gyral landmarks were then used to analyze the wiring of the system by pairing each frontal gyrus with any other temporal, parietal, and occipital gyrus of the same hemisphere. This approach overcomes limitations in previous descriptions by tackling two main issues. First, it refrains from relying on a priori definitions of the bundle for segmentation, which limits the information that can be extracted from the data (*i.e.*, you select the streamlines you are looking for based on your knowledge of the bundle). Indeed, segmentation of WM bundles based on previous anatomical knowledge often results in incomplete descriptions, potentially neglecting anatomically relevant fibers. This is particularly problematic

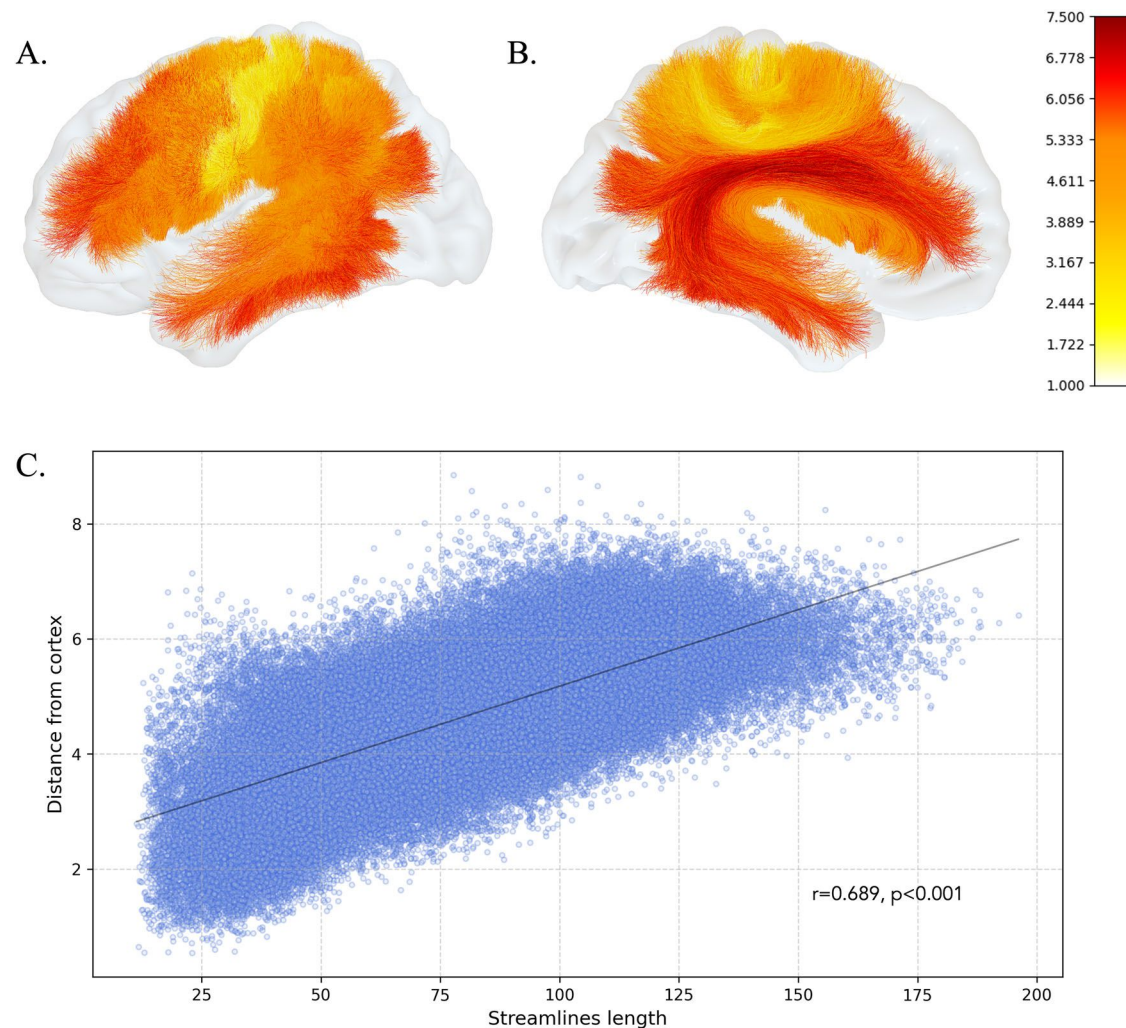


Fig. 5 | Relationship between streamline length and distance from the cortex in the SLS. Streamlines of the SLS are color-coded to reflect their distance from the cortex, with more superficial fibers shown in yellow, and deeper fibers shown in dark red (A and B, lateral and medial views, respectively). This color gradient illustrates the medio-lateral organization of the SLS, where shorter connections course more

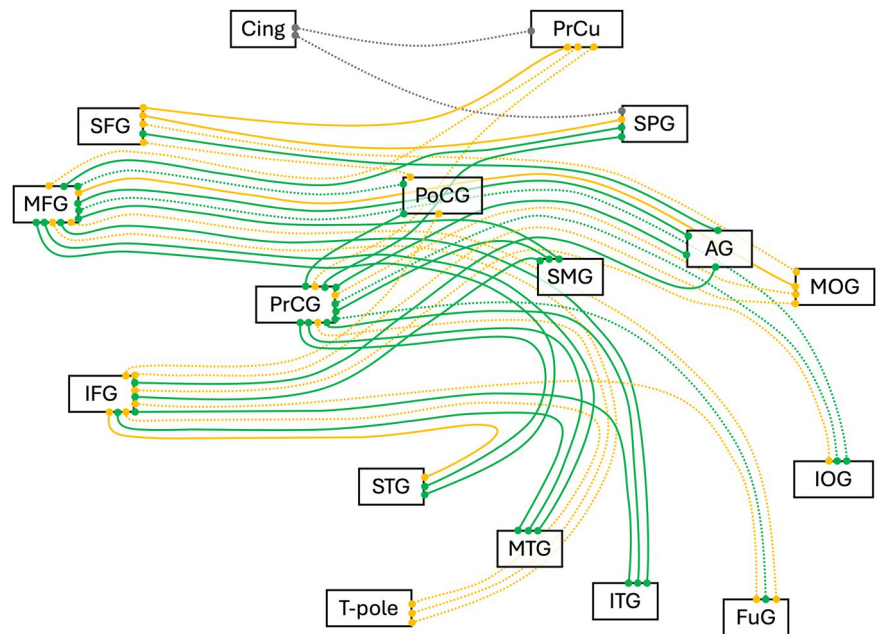
superficially, and longer connections travel deeper. The scatterplot in (C) shows a positive correlation between streamline length and distance from the cortex (Pearson's $r = 0.689$, $p < 0.001$), confirming the consistency of this organizational principle across the SLS.

considering that the anatomical definition of many WM bundles varies across different reports and is still debated even for widely studied connections such as the AF². Second, our approach does not consider WM connections as individualized entities, but rather factors in the anatomical context in which each sub-component of the system is enclosed. This allows for a meaningful depiction of the wiring of the system, revealing the actual topological organization of fibers.⁴⁶ points out the advantages of adopting a system nomenclature for describing WM pathways. However, they caution that using cortical endpoints to characterize the components of WM systems is meaningful only if these endpoints are defined by functional features. The authors argue that “*In white matter bottlenecks, streamlines intermingle and their endpoint distributions are not easily distinguishable*”, and support this claim by presenting reconstructions of part of the SLS in two HCP participants, parcellated based on either frontal or parieto-temporal terminations (see Fig. 1 in⁴⁶). By using either anterior or posterior terminations of the bundle, they highlight how fibers, after converging into the deep WM of the central region, become virtually indistinguishable in their course. In contrast, the cortex-to-cortex pairing approach adopted in the present work reveals an underlying organization of the SLS by parcellating the system based on both anterior and posterior cortical terminations. Moreover, as we integrate for the first time in-vivo tractography and ex-vivo dissection data within the same radiological space²⁵, we applied the same cortex-to-cortex

pairing strategy to the delineation of WM connections in ex-vivo dissection, providing robust anatomical validation of the sub-SLS reconstructed with tractography.

Our approach led to a comprehensive representation of the entire dorsal associative connectivity of the frontal cortex, and the innovative aspects of our integrative ex-vivo/in-vivo methodology allowed us to classify the reconstructed sub-SLS based on their anatomical plausibility and to uncover novel insights into the organization of WM fibers. As shown in Fig. 2 and in Fig. 6 and as outlined in the results section, our findings validate the existence of many connectivity patterns traditionally attributed to the SLF/AF complex (Fig. 2, sub-SLS shown in green). At the same time, a small subset of the reconstructed sub-SLS could not be confirmed by ex-vivo dissection data. This likely reflects limitations inherent to the dissection technique, such as the fixation procedure that might favor the dissection of certain fibers compared to others⁴⁷. Additionally, the latero-medial dissection approach we adopted is unsuitable for capturing more medial connections, such as SFG-SPG. Non-validated connections were further categorized as either anatomically plausible (Fig. 2, sub-SLS shown in yellow) or implausible (Fig. 2, sub-SLS shown in red) based on their spatial characteristics and relationships with the other anatomically validated sub-SLS. To contextualize our findings within the existing literature, we compare the anatomically plausible connections identified in this study with those

Fig. 6 | Schematic wiring diagram of the SLS as defined by the present study and previous literature (connectivity patterns extracted from²). Gray lines represent connections reported only in the literature, while green and yellow lines correspond to anatomically plausible sub-SLS connections identified in the current study, following the color scheme used in Fig. 2—validated in green and not validated in yellow. Solid lines denote connections found in both the literature and the present study, while dotted lines indicate connections identified in only one source. Disambiguation between literature-only and study-only connections is provided by the line color. AG angular gyrus, Cing cingulate cortex, FuG fusiform gyrus, IFG inferior frontal gyrus, IOG inferior occipital gyrus, ITG inferior temporal gyrus, MFG middle frontal gyrus, MOG middle occipital gyrus, MTG middle temporal gyrus, PoCG postcentral gyrus, PrCG precentral gyrus, PrCu precuneus, SFG superior frontal gyrus, SMG supramarginal gyrus, SOG superior occipital gyrus, SPG superior parietal gyrus, STG superior temporal gyrus, T-pole temporal pole.



previously attributed to the AF/SLF complex as summarized in our prior review². By aggregating all gyrus-to-gyrus connections previously reported, regardless of differences in nomenclature or paratomy, we define a set of similarities and differences with earlier descriptions (Fig. 6).

Our findings confirm many previously described connections between the 4 frontal gyri of the lateral convexity (i.e., IFG, MFG, SFG, and PrCG) to several parietal (i.e., AG, PoCG, SPG, and SMG), and temporal regions (i.e., ITG, MTG, and STG). However, we also identified additional plausible connections that have been either sparsely reported or not previously associated with the SLS. These include projections from the SFG, MFG, IFG, and PrCG and from MFG, IFG, and PrCG with MOG and IOG, respectively—the only connection with the occipital cortex that had previously been described in the framework of the SLS is MFG-MOG. Similarly, while earlier studies reported connections to the PrCu only from the SFG, our data-driven approach revealed plausible connections to this region also arising in the MFG and PrCG. Moreover, we identified connections to the FuG and T-pole, which had not been described in the context of the SLS. Conversely, we did not observe connections from the cingulate cortex (Cing) to the SPG and PrCu, which have been reported in previous studies. This discrepancy arises from the adoption of the system nomenclature proposed by¹, which attributes such connections to the mesial longitudinal system (MesLS) rather than the SLS. In summary, our approach unifies and extends prior work through a consistent, data-driven framework. Importantly, by reconstructing the entire wiring of the SLS with the integration of direct anatomical information, it offers further insights into the architecture of association fibers.

Indeed, the conjoint ex-vivo/in-vivo analysis uncovered fundamental organizational features of WM systems, defining plausibility principles. Not everything is connected to everything. The structural architecture of the SLS adheres to a precise and harmonic organization defined by a dorso-ventral and latero-medial axis. Our results demonstrate that connections between dorsal regions remain more dorsal, while connections between ventral regions remain more ventral (Fig. 4). Moreover, as previously described by Meyner⁴⁸, the distance between connected regions corresponds to the depth of the fibers, with shorter and more superficial fibers linking proximal areas, and longer and deeper fibers connecting distal regions. Interestingly, this organization applies to both the SLS in its entirety and to individual gyri, as shown in Fig. 5 and Supplementary Fig. 3 for tractography and in Fig. 3 for microdissection.

Of note, we incorporated the integrated in-vivo tractography/ex-vivo dissection anatomical validation into a recursive pipeline to refine the

tractography reconstruction of each sub-SLS. Initially, broad clusters computed with QuickBundlesX³² are evaluated as plausible or implausible (i.e., for acceptance or rejection) based on the geometrical features of the streamlines, following the BST templates creation process²⁶. If the merge of the tractography reconstruction preliminarily cleaned showed insufficient coverage to fully match the dissection data, we employed Tractome on the original sub-SLS reconstruction to refine the selection of plausible streamlines^{35,36}. Indeed, QuickBundlesX clustering³² can generate “mixed” clusters containing both plausible and implausible streamlines, where rejecting entire clusters risks excluding valid streamlines not represented by other accepted clusters. To address this, we used Tractome^{35,36}, supporting the iterative re-computation of a custom number of clusters, therefore enabling the selective removal of individual implausible streamlines. Reassessing each sub-SLS in light of the anatomical information gathered through the conjoint exploration of in-vivo tractography and ex-vivo dissection allowed us to generate templates that align with the plausibility of both the geometric characteristics of the streamlines and the anatomical integrity of the system. These anatomically refined tractography data will serve as an ideal source of relevant priors for the BST process, facilitating the creation of an anatomically reliable population-based atlas of the SLS²⁶.

Although the present study focuses exclusively on the structural characterization of the SLS, the anatomical insights provided here lay the foundation for future investigations into the functional and clinical relevance of these connections. By offering a comprehensive, data-driven, and anatomically validated mapping of the SLS as a multilayered WM system, we establish a framework for exploring how distributed structural connectivity supports cognitive processes. This system-level perspective aligns with the current shift in neuroscience away from the “one-bundle one-function” model and towards understanding cognition as an emergent property of integrated networks^{4,45}. Characterizing WM pathways as layered and spatially ordered systems also provides a scaffold for investigating dysfunction and recovery. In this context, our findings may inform studies of functional plasticity and reorganization of cognitive functions following brain injury, where anatomical connectivity plays a key role in shaping compensatory mechanisms.

Limitations

The number of directions and b-value used in our DWI data are not optimal for the original orientation distribution function (ODF) estimation by current standards. Indeed, the BIL&GIN data were acquired between 2010

and 2011, when the standard number of gradient direction maps in dMRI was not specifically designed for the advanced tractography methods employed today. Nevertheless, the original acquisition protocol was optimized to enhance the signal-to-noise ratio (SNR). Specifically, the DWI data were acquired with 21 gradient directions uniformly distributed across one hemisphere. These measurements were doubled by acquiring data in the 21 corresponding opposite directions in the other hemisphere, and this entire set of acquisitions was repeated twice. This approach resulted in four repetitions of each gradient direction, significantly enhancing the SNR by a factor of two through averaging. The excellent SNR of these DWI data allowed us to successfully implement fiber orientation distribution function (fODF) reconstruction using a spherical harmonic order of 6 with CSD⁴⁹. Despite using only 21 gradient directions, we obtained a robust estimation of 28 coefficients, achieving good angular contrast. Thanks to these methodological improvements in both the acquisition protocol and the diffusion modeling approach, we successfully applied advanced, cutting-edge tractography algorithms to this older dataset. Previous publications combining fODF-based tractography and microdissection validations have demonstrated the effectiveness and reliability of this methodological strategy^{26,41,42,50–54}. The replication of the tractography results obtained from the BIL&GIN DWI datasets with HCP-derived test-retest tractography confirms the reliability of the sub-SLS reconstructions provided in this study, stresses the accuracy related to the conjoint direct exploration of in-vivo tractography and ex-vivo dissection evidence, and further justifies the data-driven classification of the sub-SLS templates based on their anatomical plausibility.

In addition, although the anatomical annotation of fibers on the photogrammetric models was evaluated by three independent expert neuroanatomists, this assessment still relies on qualitative human judgment. Computer vision and machine learning approaches hold promise for automatically extracting these fiber courses. Such methods could provide objective benchmarks to complement expert evaluation. Nevertheless, given the complexity of WM anatomy and the need to interpret findings within a broader neuroanatomical context, expert anatomical assessment will remain indispensable.

While the *anatomically plausible and validated* data presented in this study provide a reliable characterization of the SLS wiring, further investigations are needed to validate the anatomical course and extension of sub-SLS whose reliability could not be investigated due to limitations related to the dissection approach adopted. Moreover, while contrasting evidence from 39 participants with multiple individual specimens sourced from different populations is the only solution currently available for integrated anatomical studies, a definitive validation of the wiring of WM pathways will be achieved only through direct integration of tractography and dissection of the same brain.

Conclusion

This study demonstrates the value of shifting from single-bundle analyses to a system perspective for the study of the structural connectivity of the brain. The adoption of a system nomenclature and the definition of the wiring of the system following the cortex-to-cortex pairing approach led to a comprehensive and data-driven representation of the wiring of the SLS that respects its topological organization. By integrating in-vivo tractography and ex-vivo dissection within a unified radiological space, we provide a novel and anatomically grounded framework for exploring the architecture of WM pathways. This approach enhances the understanding of the SLS and sets an anatomically reliable foundation for mapping the parallel and distributed functional dynamics underpinning cognitive functions.

Methods

MRI data

This study includes T1-weighted and DWI data from 39 healthy participants sourced from the BIL&GIN database, acquired on a 3T Philips Achieva²⁹. The dataset has previously been used to create BST templates of the pyramidal tract in⁵⁰, and it is included in the open-source multi-site database

TractoInferno (all males; mean age 28.1 ± 7.3 years, min 20, max 57, 8 left-handers)⁵⁵. All participants provided written informed consent, and the local ethics committee approved the study (CCPRB Basse-Normandie). All ethical regulations relevant to human research participants were followed. High-resolution 3D T1 weighted images were acquired using a 3D-FFE-TFE sequence (TR = 20 ms; TE = 4.6 ms; flip angle = 10°; inversion time = 800 ms; turbo field echo factor = 65; SENSE reduction factor = 2; field of view = $256 \times 256 \times 180$ mm; isotropic voxel = $1 \times 1 \times 1$ mm³). DWI data were acquired using a single-shot spin-echo echo-planar sequence including one b0 image ($b = 0$ s/mm²) followed by 21 diffusion gradient maps ($b = 1000$ s/mm²) distributed evenly over a half-sphere, with the 21 corresponding opposite directions spread over the opposite half-sphere. A second acquisition with identical parameters was performed to increase the signal-to-noise ratio. 70 axial slices, parallel to the AC-PC plane, were collected from the bottom of the cerebellum to the vertex with TR = 8500 ms, TE = 81 ms, flip angle = 90°, SENSE reduction factor = 2.5, field of view = 224 mm and acquisition matrix = 112×112 with $2 \times 2 \times 2$ mm³ isotropic voxels.

From whole brain tractography to sub-SLS tractograms

Whole-brain tractography was computed for each participant using CSD and particle-filtering tractography (PFT) with anatomical priors, as implemented in the TractoFlow processing pipeline^{27,28}. Default parameters were used, including a step size of 0.5 mm, an angle threshold of 20° and streamline length between 10 and 250 mm. Streamlines were seeded in the WM/gray matter (GM) interface with 10 seeds per voxel. T1-weighted images were normalized to MNI space using ANTS³⁰, and the resulting affine transformations and nonlinear deformations were applied to register the whole-brain tractograms to MNI space. Fibers of the SLS were isolated from each whole-brain tractogram through *Extractor_flow*⁵¹, leveraging the cortical gyral ROIs defined in the JHU-MNI-SS Type I White Matter Parcellation Map, which includes both cortical GM and WM labels³¹ (Fig. 1A, B). According to the anatomical definition of the SLS¹, streamlines with one termination in the frontal cortex (i.e., a concatenation of the cortical ROIs representing the PrCG, IFG, MFG, SFG, LFOG, MFOG, and RG) and one in any other gyral ROI of the same hemisphere (either parietal, occipital or temporal) were extracted. The insula served as a landmark to distinguish fibers of the SLS, which course dorsal to the insula, from fibers of the inferior longitudinal system (ILS), which pass ventrally. Streamlines ending in the cingulate cortex, thus belonging to the MeLS¹ were excluded. This process yielded 39 left and 39 right SLS tractograms in MNI space (Fig. 1B), subsequently concatenated into one left and one right N39-SLS tractogram. A cortex-to-cortex pairing approach was used to parcellate the N39-SLS left and right tractograms into sub-SLS. Hence, each ROI corresponding to a frontal gyrus was paired with any other ROI representing a parietal, occipital or temporal gyrus of the ipsilateral cortex (Fig. 1C). Streamlines connecting the two ROIs, if any, were extracted for each pair. Considering the number of frontal ($N = 7$), parietal ($N = 5$), occipital ($N = 5$), and temporal ($N = 7$) gyri included in the JHU template, a maximum of 119 possible ROI combinations were tested (Fig. 2). A shell/stem filtering approach was applied to each sub-SLS as described in Fig. 1B in⁵¹, granting the extraction of streamlines reaching the cortical surface of the gyrus after passing through its stem, thus excluding those crossing sulci. All ROI-based filtering steps leading from whole-brain tractograms to shell/stem-filtered sub-SLS, were performed with the scilpy script *scil_tractogram_filter_by_roi.py* (<https://github.com/scilpy/scilpy/tree/master/scripts>). Sub-SLS with less than 200 streamlines (i.e., less than 5 per participant on average) were discarded (Fig. 2, gray circles). This threshold was based on the tractography protocol adopted, in which each voxel at the GM–WM interface was seeded 10 times, typically generating around 5 streamlines per voxel. When streamlines from all 39 participants were aggregated to form each group-level sub-tract, this yielded an expected contribution of ~195 streamlines per voxel. We therefore used 200 streamlines as a conservative lower bound to define sub-SLS consistently represented across the population.

Building the sub-SLS templates

Symmetrical templates of each sub-SLS were created following the template creation process envisaged by the BST approach²⁶. Therefore, each right hemisphere sub-SLS was registered to the left hemisphere by flipping streamlines along the x-axis (*scil_tractogram_flip.py*) and applying an affine transformation accounting for the cerebral torque (*scil_tractogram_apply_transform.py*). We then concatenated the two homologous sub-SLS in the left hemisphere (*scil_tractogram_math.py*) and corrected for the density bias by removing duplicate streamlines (*scil_remove_similar_streamlines.py*). Streamlines of each sub-SLS were clustered with QuickBundlesX³², and clusters were evaluated based on the geometric properties of the streamlines (*scil_bundle_clean_qbx_clusters.py*) (Fig. 1D). Implausible clusters, characterized by abrupt directional changes, significant deviations from the main trajectory of the bundle, being too wiggly, proceeding further after reaching the level of their termination territories to turn back and making loops, were discarded. Mixed clusters containing both plausible and implausible streamlines were rejected at this stage. The remaining plausible streamlines were smoothed (*scil_tractogram_smooth.py*), and outliers removed (*scil_bundle_alter_to_target_dice.py*). Finalized left hemisphere templates were then flipped back to create right hemisphere homologous templates (*scil_tractogram_flip.py*).

Templates anatomical evaluation with BraDiPho

WM layer-by-layer ex-vivo microdissection dedicated to the SLS was performed in ten hemispheres (six left and four right)^{8,56,57} sourced from the Brain Anatomy Data Bank (BADaBank) of the Structural and Functional Connectivity Lab Project, Department of Neurosurgery of Santa Chiara Hospital, APSS, Trento, Italy. The study was approved by the Ethical Committee of APSS (N° 06/2018, renewed 02/24), and all ethical regulations relevant to human research participants were followed. Ex-vivo dissection and in-vivo tractography data therefore originate from separate populations. Specimens were prepared according to Klingler's method as previously described in¹⁴: after fixation in a 10% formalin solution, they were stored at -80°C for 30 days and subsequently thawed. A latero-medial dissection approach was adopted, and photogrammetry captured each dissection layer. Fully textured 3D digital models were reconstructed and registered to the T1 MRI of the specimen, as previously detailed in³³. Eight of the ten specimens used for this study are included in the BraDiPho repository of 3D human brain dissection data, openly available on the website <https://bradipho.eu>²⁵. The course of each sub-SLS was identified and annotated on the textured surface of the 3D models of each epoch following a cortex-to-cortex pairing approach corresponding to the one used for tractography data. The annotation was done manually by segmenting the point cloud of each epoch with the “Segment Tool” in the open software CloudCompare (<http://www.cloudcompare.org/>) (see Tutorial 2 at <https://bradipho.eu/2-3d-visualizer-r.html>).

Non-linear registration of the MNI T1 to the T1 of each specimen was computed in ANTS and applied to each sub-SLS template. Templates from the left hemisphere were registered to left hemisphere dissection models, while templates from the right hemisphere were registered to right hemisphere dissection models. The templates were rescaled and rendered as tubes into the photogrammetric models. The pipeline supporting the registration of fibers into each specimen's radiological space and their rendering into the photogrammetric 3D models is available at <https://zenodo.org/records/11192915>. A conjoint visual exploration of the annotations on ex-vivo dissection models of each sub-SLS and the corresponding tractography template was carried out in CloudCompare by three expert anatomists (Fig. 1E, Supplementary Fig. 1, see also “In-vivo tractography - ex-vivo dissection merging” at <https://bradipho.eu/2-3d-visualizer-r.html>). Given that the tractography sub-SLS templates were derived from a population of 39 participants, and each was evaluated against anatomical evidence from individual specimens, the procedure involved not only confirming the presence of a connection between specific cortical gyri but also assessing whether the anatomical extent of the connection as outlined in the dissection was encompassed by the corresponding tractographic

reconstruction. By virtue of being population-based, the tractography templates are expected to incorporate possible interindividual variations of the same connection.

To complement the visual assessment, we implemented a quantitative method to evaluate the spatial agreement between tractography reconstructions and annotations on the dissection epoch. This analysis involved two main steps. First, we generated a mesh capturing the regions of spatial overlap between each tractography sub-SLS template and the dissection layer displaying the corresponding WM connection. This was achieved using CloudCompare's signed distance computation tool, which produces a scalar field representing the distance of each point in the tractography template from the dissection surface. Points with a distance <1 mm were retained, forming a mesh that delineates the region of close correspondence between the dissection layer and the tractography bundle reconstruction (see Supplementary Fig. 1, third row of each panel). Next, we computed the signed distance between this overlap mesh and the manual annotation of WM fibers for the same connection, again using CloudCompare. This yielded a second scalar field on the point cloud of the manual annotation, encoding the distance of each annotated point from the region of tractography–dissection overlap (Supplementary Fig. 1, fifth row). Color-coding the manual annotation based on these distance values enabled visual identification of local discrepancies between tractography and dissection representations. To quantify agreement, we calculated the proportion of points in the annotation that lay within 1 mm of the tractography overlap mesh, yielding a percentage overlap value for each connection in different dissection epochs. These values are reported in Supplementary Data 1. For illustration, the specimens showing the highest overlap for each tract are shown in Supplementary Fig. 1.

Discrepancies between the anatomical extension of the sub-SLS template and the corresponding annotations on the dissection models prompted a refinement in the tractography template. Using the anatomical information gathered, we reclassified streamlines from the left and right sub-SLS concatenation into plausible or implausible (Fig. 1D). To achieve a more precise distinction between plausible and implausible streamlines, we employed Tractome^{35,36}, a software tool designed to iteratively refine clustering, thus supporting an accurate selection process to exclude implausible streamlines while preserving plausible ones. This approach overcomes the issue of rejecting entire mixed clusters, which might otherwise result in the unnecessary loss of valid streamlines. The refined sub-SLS templates, consisting only of plausible streamlines, were finalized following the procedures outlined in the previous paragraph and integrated into the 3D models of ex-vivo dissection for final validation.

Based on the conjoint in-vivo tractography and ex-vivo dissection analysis, sub-SLS templates were classified as *anatomically plausible and validated* (Fig. 2, green templates), *anatomically implausible but not validated* (Fig. 2, yellow templates), and *anatomically implausible* (Fig. 2, red templates). *Anatomically plausible and validated* templates are those finding correspondence in ex-vivo dissection data, authenticating both the existence of that gyrus-to-gyrus connection and its anatomical features (i.e., course and extension). The remaining templates, lacking correspondence in our ex-vivo dissection data, were classified as *anatomically plausible but not validated* or *anatomically implausible* based on whether their shape and fiber distribution exhibited coherence with the anatomical principles of fiber organization identified in the dissection data.

Templates replication study

To further assess the reliability of the tractography sub-SLS templates reconstructed from the N39 BIL&GIN DWI dataset, we performed a replication analysis on the three HCP test-retest tractography datasets released on the open platform brainlife.io, namely *test*, *test_run_2*, and *retest*³⁷. Out of the 44 participants included in these datasets, we retained 43 due to the lack of *test_run_2* tractogram for *sub-200614* in the original release. To verify the adherence of the BIL&GIN-derived sub-SLS templates with the tractography of the HCP datasets while limiting the effect of artefactual reconstructions, we matched the plausible streamlines of each left

hemisphere sub-SLS template produced in the current study (Fig. 1D) with the streamlines of individual tractograms included in the three HCP tractography datasets using the Fast Streamline Search approach³⁸. We tested for three different radii values, driving the search of streamlines matching the ones of the sub-SLS plausible templates of the current study in the HCP test-retest whole-brain tractograms within a radius of 2.0, 3.0, and 4.0 mm. The best agreement between the sub-SLS reconstructions from the different datasets in terms of spatial coverage and cortical terminations was obtained with radius=3.0. The derived individual HCP sub-SLS streamlines were then independently concatenated for each dataset (i.e., *test*, *test_run_2*, and *retest*). To quantify the adherence of the BIL&GIN-derived sub-SLS templates to those extracted from the HCP-derived datasets we considered wDSC. We calculated further streamline- and voxel-based similarity measurements, namely bundle adjacency, dice coefficient, density correlation, overlap and overreach as implemented in the *scil_bundle_pairwise_comparison.py* script⁵⁸.

SLS anatomical description

Shape analysis was performed on each finalized sub-SLS template (Supplementary Data 3)³⁹, and the mean x, y, and z coordinates of their terminations were extracted (Supplementary Data 4). A qualitative anatomical description of all templates is provided in the results section, with templates grouped based on their anterior termination. To explore the topological organization of the system, centroids of each *anatomically plausible* sub-SLS template, both *validated* and *not validated*, were computed (*scil_bundle_compute_centroid.py*). The spatial distribution of the templates was analyzed by considering both individual frontal gyri and frontal gyri altogether. As a pattern of streamlines distribution reflecting fiber length and depth of course was observed qualitatively in both dissection and tractography data, Pearson's correlation coefficient (*r*) was calculated to assess the relationship between the two measurements. Statistical analysis was performed in Python 3.10, and level for significance was set at *p* < 0.05.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

.*trk* files of the sub-SLS templates in MNI space are available at <https://doi.org/10.5281/zenodo.1675141>³⁹. *.bin* files for each specimen containing 3D digital models of layer-by-layer Klingler dissection, manual annotations of sub-SLS connections on the dissection models, and matching sub-SLS tractography templates non-linearly registered to the space of the specimen as presented in Fig. 1 and Supplementary Fig. 1 are available at <https://bradipho.eu>. These data can be visualized in the software CloudCompare (<http://www.cloudcompare.org/>). A tutorial for data browsing is available at <https://youtu.be/9WKbwgMnsNE>.

Code availability

All the scripts used for the creation of the tractography templates as outlined for each data processing step in the main text are available at <https://github.com/scilus/scilpy/tree/master/scripts>. The codes supporting the integration and rendering of in-vivo tractography data into the photogrammetric 3D digital models of ex-vivo dissection are available at <https://zenodo.org/records/11192915> and <https://github.com/minilabus/bradiphopy/tree/main/scripts>, respectively.

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References

- Mandonnet, E., Sarubbo, S. & Petit, L. The nomenclature of human white matter association pathways: Proposal for a systematic taxonomic anatomical classification. *Front. Neuroanat.* **12**, 94 (2018).
- Vavassori, L., Sarubbo, S. & Petit, L. Hodology of the superior longitudinal system of the human brain: a historical perspective, the current controversies, and a proposal. *Brain Struct. Funct.* **22**, 1363–1384 (2021).
- Thiebaut de Schotten, M., Dell'Acqua, F., Valabregue, R. & Catani, M. Monkey to human comparative anatomy of the frontal lobe association tracts. *Cortex* **48**, 82–96 (2012).
- Forkel, S. J., Friedrich, P., Thiebaut de Schotten, M. & Howells, H. White matter variability, cognition, and disorders: a systematic review. *Brain Struct. Funct.* <https://doi.org/10.1007/s00429-021-02382-w> (2021).
- Dick, A. S. & Tremblay, P. Beyond the arcuate fasciculus: Consensus and controversy in the connective anatomy of language. *Brain* **135**, 3529–3550 (2012).
- Dejerine, J. & Dejerine-Klumpke, A. Anatomie des centres nerveux. Tome 1. Paris: Rueff et Cie, (1895).
- Dejerine, J. & Dejerine-Klumpke, A. Anatomie des centres nerveux. Tome 2. Paris: Rueff et Cie, (1901).
- Agrawal, A. et al. Josef Klingler's Models of White Matter Tracts: Influences on neuroanatomy, neurosurgery, and neuroimaging: *Neurosurgery* **69**, 238–254 (2011).
- Jeurissen, B., Descoteaux, M., Mori, S. & Leemans, A. Diffusion MRI fiber tractography of the brain. *NMR Biomed.* **32**, e3785 (2019).
- Catani, M., Jones, D. K. & ffytche, D. H. Perisylvian language networks of the human brain. *Ann. Neurol.* **57**, 8–16 (2005).
- Glasser, M. F. & Rilling, J. K. DTI Tractography of the Human Brain's Language Pathways. *Cereb. Cortex* **18**, 2471–2482 (2008).
- Makris, N. et al. Segmentation of subcomponents within the superior longitudinal fascicle in humans: A quantitative, in vivo, DT-MRI study. *Cereb. Cortex* **15**, 854–869 (2005).
- Fernández-Miranda, J. C. et al. Three-dimensional microsurgical and tractographic anatomy of the white matter of the human brain. *Neurosurgery* <https://doi.org/10.1227/01.NEU.0000297076.98175.67> (2008).
- Sarubbo, S. et al. Structural and functional integration between dorsal and ventral language streams as revealed by blunt dissection and direct electrical stimulation: Anatomico-Functional Integration of Language. *Hum. Brain Mapp.* **37**, 3858–3872 (2016).
- Zacà, D. et al. Whole-brain network connectivity underlying the human speech articulation as emerged integrating direct electric stimulation, resting state fMRI and tractography. *Front. Hum. Neurosci.* **12**, 405 (2018).
- De Benedictis, A. et al. Anatomico-functional study of the temporo-parieto-occipital region: dissection, tractographic and brain mapping evidence from a neurosurgical perspective. *J. Anat.* **225**, 132–151 (2014).
- Martino, J. et al. Cortex-sparing fiber dissection: an improved method for the study of white matter anatomy in the human brain: Cortex-sparing fiber dissection. *J. Anat.* **219**, 531–541 (2011).
- Fernández-Miranda, J. C. et al. Asymmetry, connectivity, and segmentation of the arcuate fascicle in the human brain. *Brain Struct. Funct.* **220**, 1665–1680 (2015).
- Frey, S., Campbell, J. S. W., Pike, G. B. & Petrides, M. Dissociating the human language pathways with high angular resolution diffusion fiber tractography. *J. Neurosci.* **28**, 11435–11444 (2008).
- Vassal, F. et al. Combined DTI tractography and functional MRI study of the language connectome in healthy volunteers: Extensive mapping of white matter fascicles and cortical activations. *PLoS ONE* **11**, e0152614 (2016).
- de Schotten, T. & Thiebaut M. Direct evidence for a parietal-frontal pathway subserving spatial awareness in humans. *Science* **309**, 2226–2228 (2005).
- Martino, J. et al. Fiber dissection and diffusion tensor imaging tractography study of the temporo-parietal fiber intersection area. *Oper. Neurosurg.* **72**, ons87–ons98 (2013).

23. Forkel, S. J. et al. Anatomical evidence of an indirect pathway for word repetition. *Neurology* **94**, e594–e606 (2020).
24. Pestilli, F., Yeatman, J. D., Rokem, A., Kay, K. N. & Wandell, B. A. Evaluation and statistical inference for human connectomes. *Nat. Methods* **11**, 1058–1063 (2014).
25. Vavassori, L. et al. Brain dissection photogrammetry for studying human white matter connections: A unique resource for integrating ex-vivo and in-vivo multimodal datasets. Preprint at <https://doi.org/10.21203/rs.3.rs-6480729/v1> (2025).
26. Rheault, F. et al. Bundle-specific tractography with incorporated anatomical and orientational priors. *NeuroImage* **186**, 382–398 (2019).
27. Theaud, G. et al. TractoFlow: A robust, efficient and reproducible diffusion MRI pipeline leveraging Nextflow & Singularity. *NeuroImage* **218**, 116889 (2020).
28. Girard, G., Whittingstall, K., Deriche, R. & Descoteaux, M. Towards quantitative connectivity analysis: reducing tractography biases. *NeuroImage* **98**, 266–278 (2014).
29. Mazoyer, B. et al. BIL&GIN: A neuroimaging, cognitive, behavioral, and genetic database for the study of human brain lateralization. *NeuroImage* **124**, 1225–1231 (2016).
30. Avants, B. B., Tustison, N. & Johnson, H. Advanced Normalization Tools (ANTS). 2, 1–35 (2009).
31. Oishi, K. et al. Atlas-based whole brain white matter analysis using large deformation diffeomorphic metric mapping: Application to normal elderly and Alzheimer's disease participants. *NeuroImage* **46**, 486–499 (2009).
32. Garyfallidis, E., Côté, M.-A., Rheault, F. & Descoteaux, M. QuickBundlesX: Sequential clustering of millions of streamlines in multiple levels of Figures detail at record execution time. in *24th International Society of Magnetic Resonance in Medicine (ISMRM)* (2016).
33. De Benedictis, A. et al. Photogrammetry of the human brain: A novel method for three-dimensional quantitative exploration of the structural connectivity in neurosurgery and neurosciences. *World Neurosurg.* **115**, e279–e291 (2018).
34. Yagmurlu, K., Middlebrooks, E. H., Tanriover, N. & Rhoton, A. L. Fiber tracts of the dorsal language stream in the human brain. *JNS* **124**, 1396–1405 (2016).
35. Porro-Muñoz, D. et al. Tractome: A visual data mining tool for brain connectivity analysis. *Data Min. Knowl. Discov.* **29**, 1258–1279 (2015).
36. Sarubbo, S. et al. Changing the paradigm for tractography segmentation in neurosurgery: Validation of a streamline-based approach. *Brain Sci.* **14**, 1232 (2024).
37. Hayashi, S. et al. brainlife.io: a decentralized and open-source cloud platform to support neuroscience research. *Nat. Methods* **21**, 809–813 (2024).
38. St-Onge, E., Garyfallidis, E. & Collins, D. L. Fast streamline search: An exact technique for diffusion MRI tractography. *Neuroinform* **20**, 1093–1104 (2022).
39. Yeh, F.-C. Shape analysis of the human association pathways. *NeuroImage* **223**, 117329 (2020).
40. Zhang, Y. et al. Atlas-guided tract reconstruction for automated and comprehensive examination of the white matter anatomy. *NeuroImage* **52**, 1289–1301 (2010).
41. Hau, J. et al. Cortical terminations of the inferior fronto-occipital and uncinate fasciculi: Anatomical stem-based virtual dissection. *Front. Neuroanat.* **10**, 58 (2016).
42. Hau, J. et al. Revisiting the human uncinate fasciculus, its subcomponents and asymmetries with stem-based tractography and microdissection validation. *Brain Struct. Funct.* **222**, 1645–1662 (2017).
43. Warrington, S. et al. XTRACT - Standardised protocols for automated tractography in the human and macaque brain. *NeuroImage* **217**, 116923 (2020).
44. Zigiottio, L. et al. Segregated circuits for phonemic and semantic fluency: A novel patient-tailored disconnection study. *NeuroImage: Clin.* **36**, 103149 (2022).
45. Thiebaut de Schotten, M. & Forkel, S. J. The emergent properties of the connected brain. *Science* 505–510 <https://doi.org/10.1126/science.abq2591> (2022).
46. Van Den Hoven, E. et al. Time to bury the chisel: A continuous dorsal association tract system. *Brain Struct. Funct.* **229**, 1527–1532 (2024).
47. Zemmoura, I. et al. FIBRASCAN: A novel method for 3D white matter tract reconstruction in MR space from cadaveric dissection. *NeuroImage* **103**, 106–118 (2014).
48. Meynert, T. *Psychiatrie: Klinik Der Erkrankungen Des Vorderhirns Begründet Auf Dessen Bau, Leistungen Und Ernährung.* (Wilhelm Braumüller, 1884).
49. Descoteaux, M., Deriche, R., Knosche, T. R. & Anwander, A. Deterministic and probabilistic tractography based on complex fibre orientation distributions. *IEEE Trans. Med. Imaging* **28**, 269–286 (2009).
50. Chenot, Q. et al. A population-based atlas of the human pyramidal tract in 410 healthy participants. *Brain Struct. Funct.* **224**, 599–612 (2019).
51. Petit, L. et al. The structural connectivity of the human angular gyrus as revealed by microdissection and diffusion tractography. *Brain Struct. Funct.* **228**, 103–120 (2023).
52. De Benedictis, A. et al. New insights in the homotopic and heterotopic connectivity of the frontal portion of the human corpus callosum revealed by microdissection and diffusion tractography: Homo- and Hetero-Topic Fronto-Callosal Connectivity. *Hum. Brain Mapp.* **37**, 4718–4735 (2016).
53. Legarreta, J. H. et al. Filtering in tractography using autoencoders (FINTA). *Med. Image Anal.* **72**, 102126 (2021).
54. Legarreta, J. H., Petit, L., Jodoin, P.-M. & Descoteaux, M. Generative Sampling in Bundle Tractography using Autoencoders (GESTA). *Med. Image Anal.* **85**, 102761 (2023).
55. Poulin, P. et al. TractoInferno: A large-scale, open-source, multi-site database for machine learning dMRI tractography. *Sci. Data* **9**, 725 (2022).
56. Ludwig, E. & Klingler, J. Atlas cerebri humani: Der innere Bau des Gehirns dargestellt auf Grund makroskopischer Präparate. The inner structure of the brain demonstrated on the basis of macroscopical preparations. Little, Brown, (1956).
57. Dziedzic, T. A., Balasa, A., Jeżewski, M. P., Michałowski, Ł. & Marchel, A. White matter dissection with the Klingler technique: a literature review. *Brain Struct. Funct.* **226**, 13–47 (2021).
58. Schilling, K. G. et al. Tractography dissection variability: What happens when 42 groups dissect 14 white matter bundles on the same dataset? *NeuroImage* **243**, 118502 (2021).
59. Vavassori, L. Superior Longitudinal System (SLS) Tractography Templates. Zenodo (2025).

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Competing interests

The authors declare no competing interests.

Additional information

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